

VQE callback with gradient in some cases only records a single iteration

<https://github.com/Qiskit/qiskit/issues/8726>

ROW 42

VQE callback with gradient in some cases only records a single iteration #8726

New issue



Closed as not planned



jiayu-shen opened on Sep 10, 2022 · edited by jiayu-shen

Edits ...

Environment

- Qiskit version: 0.37.2
- Qiskit Terra version: 0.21.2
- Qiskit Aer version: 0.10.4
- Python version: 3.9.0
- Operating system: macOS Big Sur 11.6.4

What is happening?

When gradient is passed into VQE, with some combinations of `grad_method` and the optimizer, only a single iteration is passed into the callback function of VQE.

How can we reproduce the issue?

The code is modified from https://qiskit.org/documentation/tutorials/algorithms/02_vqe_convergence.html.

```
import numpy as np
import pylab

from qiskit import Aer
from qiskit.opflow import X, Z, I
from qiskit.utils import QuantumInstance, algorithm_globals
from qiskit.algorithms import VQE, NumPyMinimumEigensolver
from qiskit.algorithms.optimizers import COBYLA, L_BFGS_B, SLSQP, ADAM, CG, SPSA
from qiskit.circuit.library import TwoLocal

from qiskit.opflow.gradients import Gradient

H2_op = (-1.052373245772859 * I ^ I) + \
        (0.39793742484318045 * I ^ Z) + \
        (-0.39793742484318045 * Z ^ I) + \
        (-0.01128010425623538 * Z ^ Z) + \
        (0.18093119978423156 * X ^ X)

algorithm_globals.random_seed = 50
ansatz = TwoLocal(rotation_blocks='ry', entanglement_blocks='cz')

optimizer = SLSQP(maxiter=60)

counts = []
values = []
def store_intermediate_result(eval_count, parameters, mean, std):
    counts.append(eval_count)
    values.append(mean)

vqe = VQE(ansatz, optimizer, callback=store_intermediate_result,
          gradient=Gradient(grad_method='fin_diff'),
          quantum_instance=QuantumInstance(backend=Aer.get_backend('statevector_simulator')))
result = vqe.compute_minimum_eigenvalue(operator=H2_op)
print(f'Value using Gradient: {result.eigenvalue.real:.5f}')
```

The last plot on the original page does not display a point, but if we plot the callback result with the marker set explicitly, the single point shows up.

```
pylab.rcParams['figure.figsize'] = (12, 8)
pylab.plot(counts, values, label=type(optimizer).__name__, marker='.')
pylab.xlabel('Eval count')
pylab.ylabel('Energy')
pylab.title('Energy convergence using Gradient')
pylab.legend(loc='upper right');
```

Assignees

No one assigned

Labels

bug mod: algorithms

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Notifications

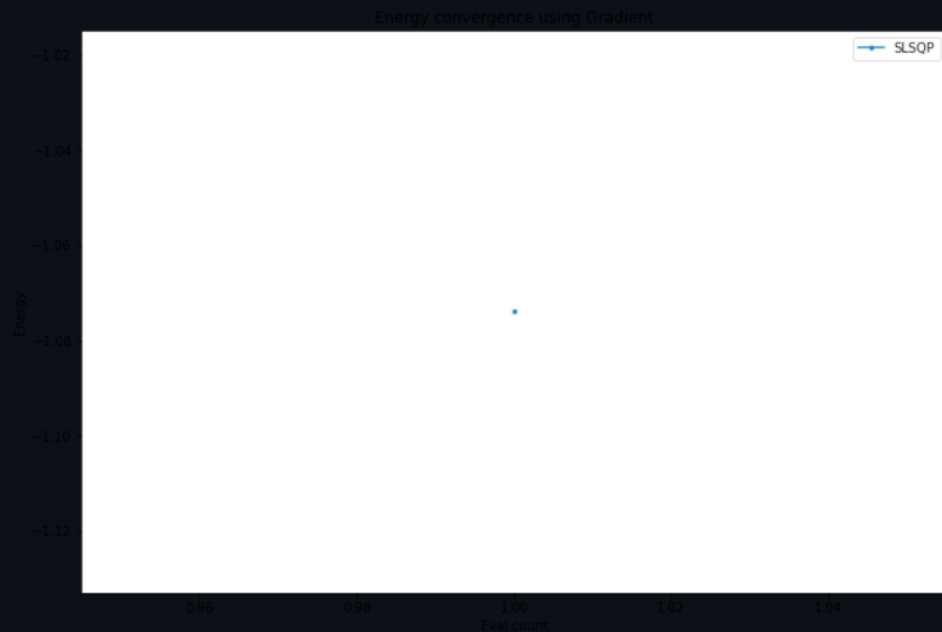
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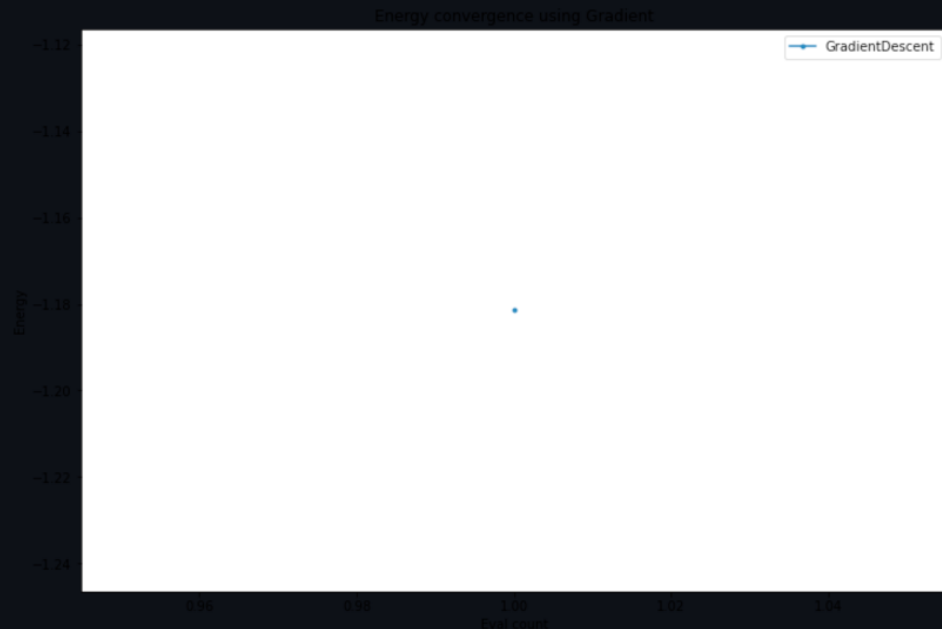


This plot is supposed to show a history in optimization, but only a single iteration is passed into the callback. This single value is probably the very first iteration. The ipynb file https://github.com/Qiskit/qiskit-tutorials/blob/master/tutorials/algorithms/02_vqe_convergence.ipynb, however, did display a plot of history.

The same issue occurs when only the optimizer is changed to ADAM `optimizer = ADAM(maxiter=60, lr=0.01)` still with `gradient=Gradient(grad_method='fin_diff')`, the output is a single point with Value using Gradient: -1.07393. With ADAM `optimizer = ADAM(maxiter=60, lr=0.01)` and `gradient=Gradient(grad_method='param_shift')` in VQE, the code takes longer to run, and produces still a single point but at a lower value Value using Gradient: -1.81570.

When using

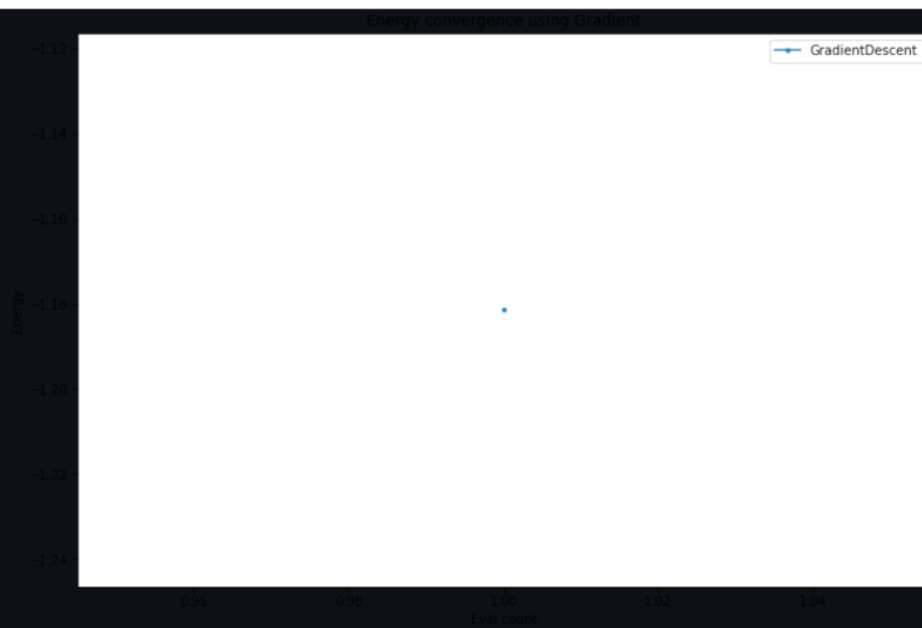
`NaturalGradient(grad_method='param_shift', qfi_method='lin_comb_full', regularization='ridge')` as the gradient in VQE and `optimizer = GradientDescent(maxiter=10)`, there is still a single point at -1.181444027905341, as is shown in the plot below



Changing from `grad_method='param_shift'` to `grad_method='lin_comb'` has a similar issue.

What should happen?

For the cases of the original tutorial page, "ADAM + Gradient(`grad_method='param_shift'`)", and "GradientDescent + NaturalGradient", the callback is expected to store the history of optimization.



Changing from `grad_method='param_shift'` to `grad_method='lin_comb'` has a similar issue.

What should happen?

For the cases of the original tutorial page, "ADAM + Gradient(`grad_method='param_shift'`)", and "GradientDescent + NaturalGradient", the callback is expected to store the history of optimization.

Any suggestions?

No response



jiayu-shen added **bug** on Sep 10, 2022



woodsp-ibm added **mod: algorithms** on Sep 11, 2022



woodsp-ibm on Jul 11, 2023

Member ...

This is for the older QuantumInstance based VQE. The new VQE uses the new gradients. The older VQE is now deprecated and soon to be removed. As such I am going to close this as at this stage, even if the problem still exists, its not going to get addressed.



woodsp-ibm closed this as **not planned** on Jul 11, 2023